EFFECTIVE SURVEYS FOR RESEARCH

M&M Postgrad Research Methodology – Prof Deborah Blaine
Recorded Presentation – April 2020

Surveys are easy, right?…
An **effective** survey requires **careful design**

- Careful and thoughtful design of surveys is necessary to create GOOD QUALITY, HIGH VALUE DATA
- There is a CODE OF ETHICS that must be followed to ensure honesty, respect and integrity when dealing with the respondents and the public
- Respondents must know the following information in order to make an INFORMED DECISION about participating in the survey
  - What will be covered in the survey (CONTENT)?
  - Who is benefiting from the survey (SPONSORSHIP)?
  - What is the purpose of the survey (PURPOSE)?
  - How will confidentiality and anonymity be ensured (PROTECTION OF DATA)?
- The researcher should be able to clearly convey the LIMITATIONS and SHORTCOMINGS of the data to the sponsors

Different types of survey research

Source: Kelly, Clarke et al, 2003, Int J of Qual in Health Care, 15:3

Descriptive	Analytical studies	Evaluative
Observe/gather information on phenomena	Look deeper into the phenomena, to reveal problems or issues	Determine the effects of change
Describe important features relevant to the situation, e.g. demographics, socio-economic aspects, geographical aspects	Conduct data analysis in order to determine the effect of one set of influences on another	

Requires statistical analysis, sufficient sample size, strong understanding of probability and statistical theories

Cornerstone of good survey research: **Well designed Research Question**

- Define a single clear and explicit research question (overarching)
- How to find your research question?
  - **Know your research area well:** independent learning, reading
  - **Widen your experience:** read widely, explore related fields, speak to experts and colleagues in the field
  - **Use techniques to bring creativity to your study:** brainstorming, discussions
  - **Avoid pitfalls:** posing unanswerable questions, asking questions that have already been answered

Is survey research the correct approach?

Yes

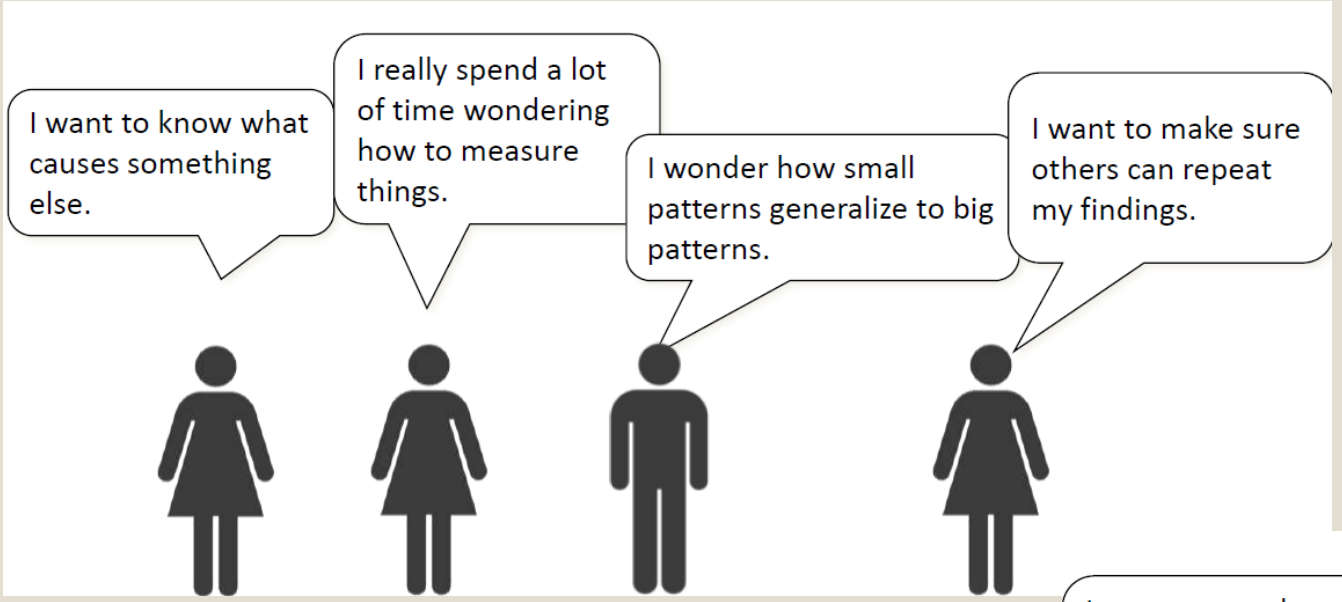
- I know a lot about the topic
- I need to summarize the findings in numerical format
- I don't need a lot of detail and nuance
- I need to generalize the results to the population
- I need to predict the likelihood of a certain thing happening
- I want to measure incremental change

No

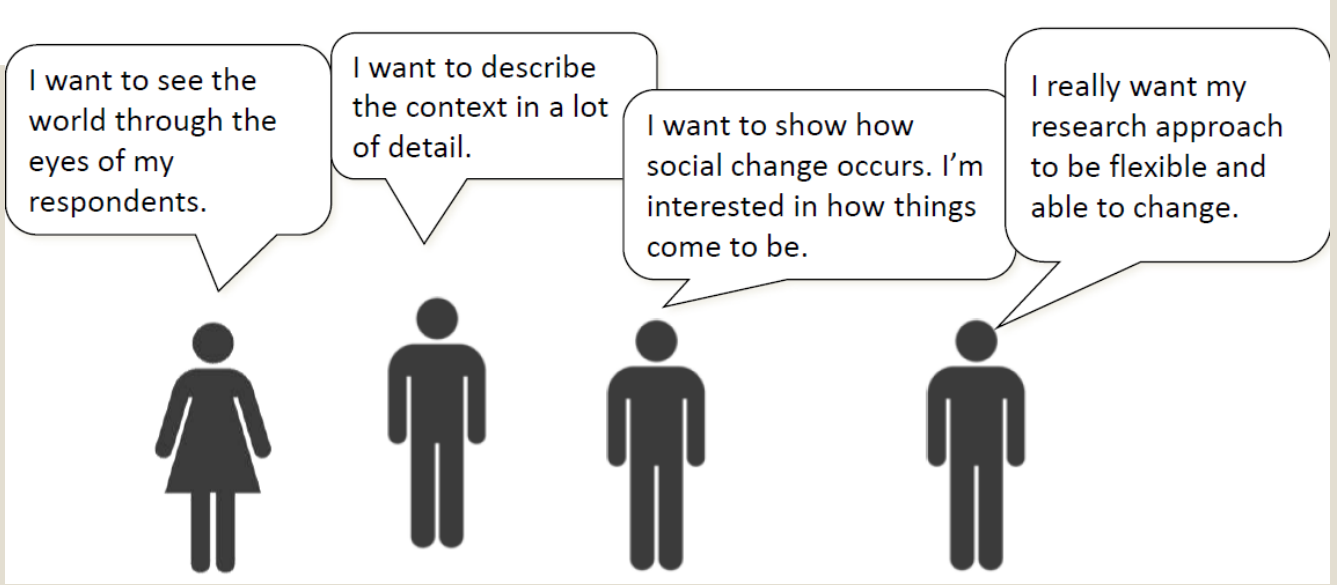
- I don't really know much about the topic
- I don't need to summarize the findings in numerical format
- I need to communicate a lot of detail and nuance
- I'm interested in case study information
- I'm not planning on predicting the likelihood of anything
- I don't need to measure any change

Source: https://www.slideshare.net/sladner/survey-research-in-design [Accessed: 23 March 2020]

Quantitative vs

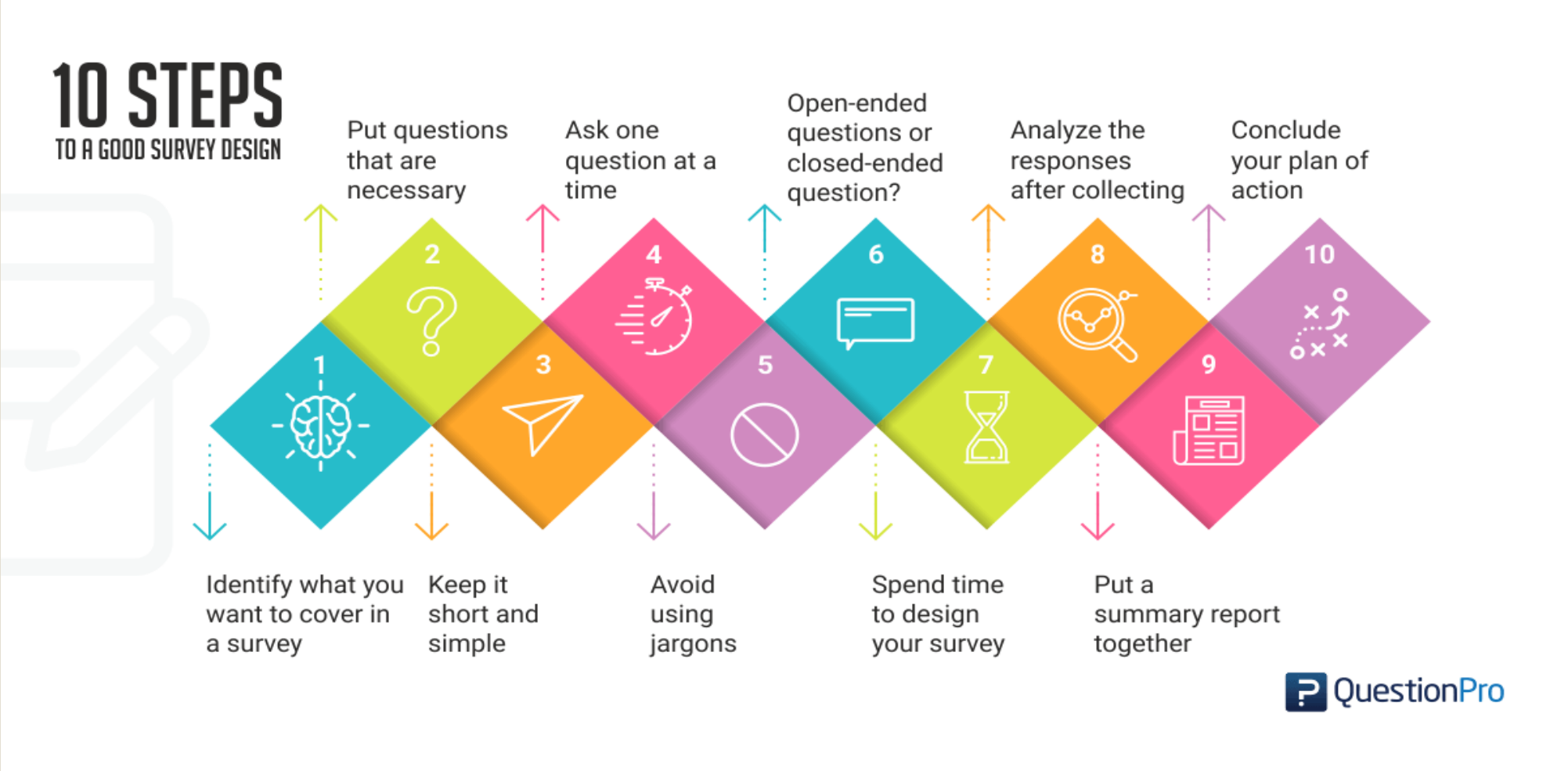


Qualitative research



Survey design

Source: https://www.questionpro.com/features/survey-design/ [Accessed 23 March 2020]



Survey mode

- Mail/e-mail – hardcopy or PDF questionnaire
- Phone interview
- In-person interview
- Online/electronic survey

Considerations

- Appearance – user friendly
- Clear instructions and questions
- If reading out the question (interviews), provide the possible answers as well as an “I don’t know” or “Not applicable” option

Survey questionnaire good practice:

- Easy to read, clear questions
- Questions should be numbered and grouped in topics
- Clear instructions and headings make the questionnaire easier to follow
- Short questionnaires are more appealing than long
- Avoid ambiguous or double-barrel questions (e.g. Are you finding this course useful and the lecture style effective?)

The ideal question accomplishes three goals:

It measures the underlying concept it is intended to tap

It doesn't measure other concepts

It means the same thing to all respondents

How to build a solid questionnaire…

Avoid technical jargon

- How effective is the dampening effect of the shock adsorbers?

  vs How smooth is the ride on a bumpy road?

Avoid vague/imprecise terms

- Which car features are important to you?

  vs Rate the following features of a car in order of importance:

  - ❑ Safety features (airbags, ABS, safety rating, ISOFIX)
  - ❑ Convenience features (window defoggers, keyless entry, entertainment system, cameras);
  - ❑ Suspension (road handling and ride quality);
  - ❑ Engine size (power, fuel efficiency)

Be very specific

- Do you service your car?

  vs Do you follow the recommended service schedule for your car?

How to build a solid questionnaire…

Avoid complex sentences

- Do you think that the oil price directly or indirectly affects the rise in car prices in South Africa, year on year?

Provide reference frames

- How important is the oil price to the industry?
- How important is the oil price to the automotive industry in South Africa currently?

Provide clearly differentiated scales

- How important is the oil price to the automotive industry in South Africa currently?
  - (1) Irrelevant; (2) Not so important; (3) Insignificant; (4) Relevant; (5) Very Important
  - (1) Irrelevant; (2) Somewhat important; (3) Relevant; (4) Important; (5) Critical

How to build a solid questionnaire…

Anticipate all possible answers

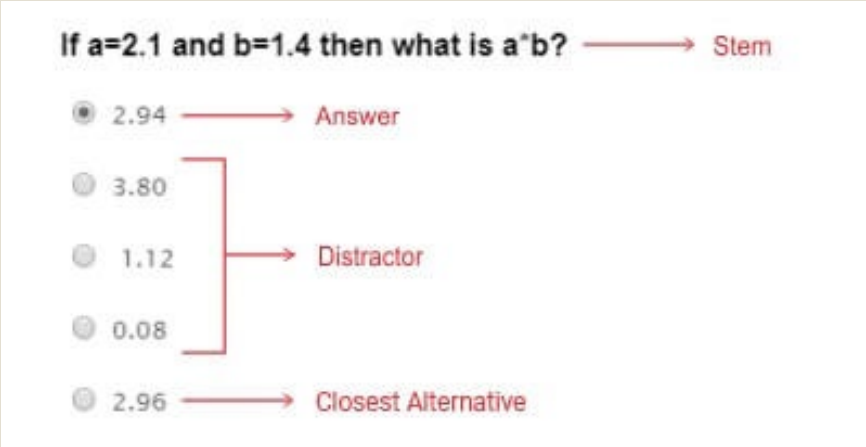
- Rate the following features of a car in order of importance:
  - ❑ Safety features (airbags, ABS, safety rating, ISOFIX)
  - ❑ Convenience features (window defoggers, keyless entry, entertainment system, cameras);
  - ❑ Suspension (road handling and ride quality);
  - ❑ Engine size (power, fuel efficiency)
  - ❑ Other: specify_____

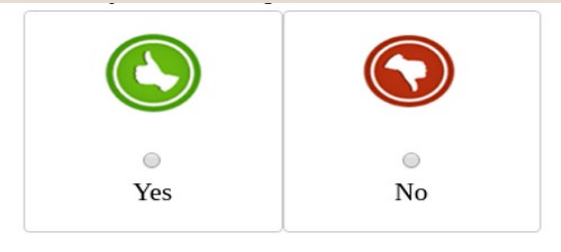
Avoid leading or emotive questions

- Do you think that frivolous features, such as a top sound system, are important when buying a car?

Open-ended vs Closed-ended questions

Close Ended Questions	Open Ended Questions
Do you like working with us? ❑ Yes ❑ No	Tell us about your experience with our organization so far.
Have you been stressed lately? ❑ Yes ❑ Unsure ❑ No	Share with us what has been troubling you.
How satisfied are you with your current job role? ❑ Very satisfied ❑ Somewhat satisfied ❑ Somewhat unsatisfied ❑ Very unsatisfied	What do you expect from this appraisal?





Advantages/Disadvantages of open/closed ended questions

- Open-ended questions allow the greatest variety of responses, but are time consuming to ask and require a lot of work to analyse
- Open-ended questions should be used where the possible replies are unknown or too numerous
- Closed ended questions, when well designed, ensure that respondents interpret questions the same way.
- Respondents are more likely to skip an open-ended than closed-ended question

Critical design features for surveys

Spend time designing your survey, considering the following features critically:

- Design an effective & efficient survey
  - *Keep it short; remove redundant, irrelevant, boring questions*
- Question order
  - *Provide an short introduction and descriptive titles for sections – this helps with the flow and keeps respondents engaged*
  - *Start with general questions that are easy to answer – help the respondent to become engaged and think about the topic*
  - *Be aware of how previous questions may influence the mood and response of subsequent questions – plan the order carefully*
  - *Leave sensitive questions, including questions regarding demographics (gender, age, income) until last – they distract and influence answers*
- Rating/Likert scales
  - *Be aware that people are biased to "agree" with statements – create a balance of positively (a positive response relates to agree) and negatively (a negative response relates to agree) stated questions*

Ethics: SU Research Ethics Committees (REC)

Research Ethics Committee: Social, Behavioural and Education Research (REC: SBE)

- non-medical, human research and the processing of personal information for the purpose of research

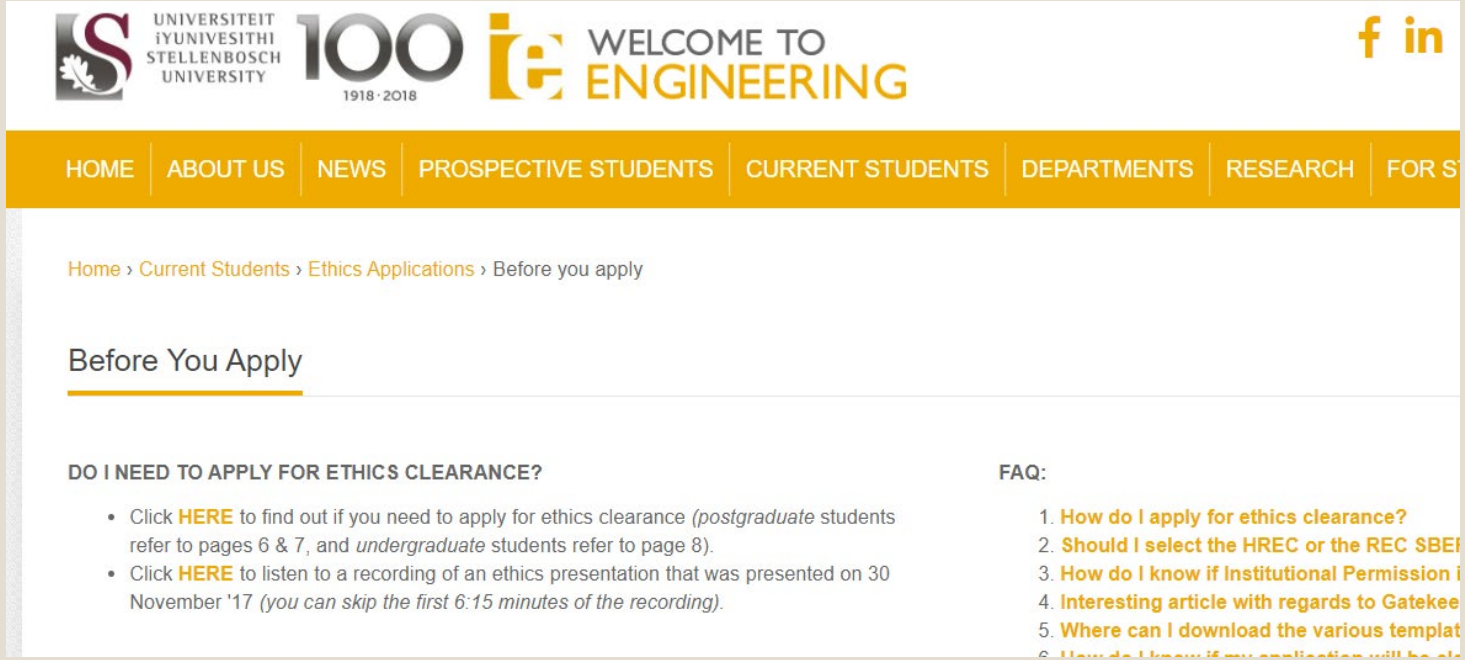
Health Research Ethics Committees (HREC1 and HREC2)

- biomechanical research, or research that involve the use of biomedical technology and applications, or the testing of mobility aids;
- research where blood or tissue samples will be collected or analysed;
- research targeting persons diagnosed with a medical, psychiatric or physical condition, where the purpose of the research is to study the condition itself and/or test interventions that could improve the condition or the quality of life of the participant;
- research that involves interaction with or the observation of patients in hospitals or other healthcare settings;
- research that involve the participation of healthy adults or children in physical or exercise activities that could result in injury[1];
- research that involves the testing and consumption of products, drinks or food stuffs by human panel members which have been enhanced by additives, not usually found in the food product
- the retrospective review and analysis of identifiable patient and/or other medical records.

Any research that includes interaction with people, including surveys, requires ethical clearance

Additional considerations for ethics application

http://www.eng.sun.ac.za/current-students/ethics-applications/before-you-apply/



Determining the level of risk: minimal – low – medium – high
http://www.eng.sun.ac.za/media/sites/7/Determining-the-Level-of-Risk.pdf

Documents required for submitting your ethics application for review

(Instructions here http://www.sun.ac.za/english/research-innovation/Research-Development/Pages/REC-Documents.aspx)

1. Research proposal
   - Research aim & Research question
   - Motivation
   - Will the research include human participants?
     - How will the participants be selected? Which participants? Why?
     - What are all the ethical considerations/risks? What risk mitigation steps are considered?
   - Methodology and data collection (these are NOT the same!)
     - e.g. design review of product with online survey to assess suitability of the design changes
2. Data collection instruments
   - Interview questions, online survey questions, video recording setups etc
3. Consent forms
   - Informed consent (assent in case of minors) forms that will be signed by participants
   - Templates are available
4. Information sheet
   - Detailed information on research study that participants can take home and read in their own time
5. Institutional clearance
   - Permission letter from institution stating that the research may access private information of the participants OR
   - Letter showing that the research has applied for institutional clearance and is awaiting the outcome.
6. Recruitment material
   - Any flyers/correspondence/posters etc used to recruit participants

SU Online ethics application process

http://www.sun.ac.za/english/research-innovation/Research-Development/Pages/Electronic-Application-Process.aspx

Not secure | sun.ac.za/english/research-innovation/Research-Development/Pages/Electronic-Application-Process.aspx

Recipes Crafts SU SharePoint Gmail Google Calendar Google Drive Google Scholar Pharos Woordeboe... Standard Bank Facebook PPS P

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Electronic Application Process

The REC: Social, Behavioural and Education Research uses an electronic ethics review management system, Infonetica©, to manage the ethics application and review process. Please read the information carefully before accessing the system.

HOW TO APPLY:

REC APPLICATION PORTAL: https://applyethics.sun.ac.za

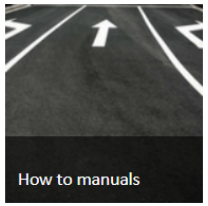
See below tabs to access instructions manuals and podcasts (videos) or to consult our Frequently Asked Questions.

Still need help?
Email or call the REC helpdesk officer, Mr Aden Williams: applyethics@sun.ac.za / 021 808 9185

OR **Click here** for the contact details of other persons who can assist you.

PLEASE NOTE:

1. Only registered SU students and staff with a valid username and password can log in to the REC portal.



Summary

- Designing an effective survey requires time and thought – you need to
  - educate yourself on the research topic by reading and discussing with other researchers
  - make sure your overarching research question is relevant and well-designed
  - Decide who your participants should be and how you will recruit them
  - Decide how you are going to analyse the data and design appropriate questions
  - Plan the type of questions: open-ended or close-ended (yes/no; multiple choice; Likert scale)
  - Plan the order of questions taking into consideration the influence of the position of questions on the respondent
  - Create clear instructions and group questions into clear sections with titles
  - Make sure that there are no repetitive, confusing or redundant questions
  - Perform a pilot run and fine-tune the survey before finalising it (this may require several iterations, gathering feedback from peers may be helpful)
  - Prepare an ethics application (including research proposal, consent forms etc)
- Administering the survey, collecting the data and analysing the results takes a significant amount of time – you to need to plan for this too

Reference documents and websites

- QuestionPro commercial survey software blog: https://qa.questionpro.com/blog/steps-to-a-good-survey-design/
- Kate Kelley, Belinda Clark, Vivienne Brown, John Sitzia, **Good practice in the conduct and reporting of survey research.** *International Journal for Quality in Health Care*, Volume 15, Issue 3, May 2003, Pages 261–266, https://doi.org/10.1093/intqhc/mzg031
- Harvard University Program on Survey Research: Questionnaire Design Tip Sheet: https://psr.iq.harvard.edu/book/questionnaire-design-tip-sheet
- Using Surveys in Design by Sam Ladner, 2010 lecture at Ontario College of Art and Design: https://www.slideshare.net/sladner/survey-research-in-design